

Prithvi Narayan Campus
Tribhuvan University
Institute of Science and Technology



Project Proposal on
Sikshya Kendra (Student Portal)

Submitted To:

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Submitted By:

Milan Shahi (30047/078)

Sandesh Subedi (30069/078)

Sugam Pokharel (30078/078)

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Abstract:

Sikshya Kendra is a modern student portal designed to centralize academic resources and simplify information access for students across various educational institutions. Unlike traditional institute specific systems, Sikshya Kendra offers a generalized platform where students can seamlessly access essential academic materials such as syllabi, notices, notes, and old question papers. Developed using HTML, CSS, and JavaScript for the frontend, and powered by Django (Python) on the backend, the system ensures a dynamic and secure user experience. The portal is equipped with intuitive navigation, responsive design, and an efficient database management system using SQLite and MySQL. It also features an admin validation system for student registration to ensure data integrity. By addressing the lack of accessible academic content platforms in the educational ecosystem, Sikshya Kendra aims to enhance self-learning, academic collaboration, and overall student engagement. It provides a user friendly and reliable digital solution for students to manage their academic journey more effectively.

Keywords: Student Portal, Academic Resources, Django, HTML/CSS/JS, SQLite, MySQL, Self-Learning, Educational Platform

Table of Contents

1. Introduction.....	1
2. Problem Statement.....	2
3. Objectives	3
4. Scope.....	4
5. Methodology	5
5.1 Requirement Identification.....	6
5.1.1 Literature Review	6
5.1.2 Study of Existing Systems.....	7
5.1.3 Requirement Analysis	8
5.2 Feasibility Study.....	9
5.3 High Level design of system.....	11
5.3.1 Use case diagram.....	11
5.3.2 System Flow Diagram	12
6. Expected Outcomes	13
7. References.....	14

List of Figures

Figure i: Gantt Chart	10
Figure ii: Use Case Diagram	11
Figure iii: System Flow Diagram	12

1. Introduction

Sikshya Kendra is a centralized student portal developed to address the growing need for a unified academic platform that supports students from multiple educational institutions. Unlike traditional systems that are specific to individual schools or colleges, Sikshya Kendra functions as a general purpose academic hub where users can access syllabi, class notes, old question papers, and official notices from a single platform.

As the adoption of digital tools in education expands, there is a rising demand for platforms that are both accessible and flexible. While existing systems often focus on internal administrative tasks, Sikshya Kendra provides a common digital environment designed to benefit students irrespective of their institutional affiliation. Developed using modern web technologies such as HTML, CSS, JavaScript, Django, and SQLite, the portal ensures a secure, interactive, and responsive user experience.

By offering a scalable and intuitive system, Sikshya Kendra seeks to close the gap between scattered academic content and students in need of reliable educational resources. The platform contributes to improving academic accessibility and encourages self-directed learning across Nepal.

2. Problem Statement

Students across various institutions often face difficulties in accessing organized and verified academic materials due to the absence of a unified digital platform. Currently, there is lack of comprehensive system that serves as a common academic hub for learners from different schools or colleges. As a result, students frequently rely on unstructured sources such as social media, informal chat groups, or scattered websites to obtain study resources and academic updates. This fragmented approach not only lacks credibility but also leads to inconsistencies in academic preparation. Furthermore, most institutional portals are either limited in functionality or accessible only to specific student groups, hindering collaboration and efficient information sharing beyond institutional boundaries. These limitations contribute to outdated material circulation, communication gaps, and missed opportunities for academic engagement among students.

3. Objectives

The primary objectives of Sikshya Kendra are as follows:

- To develop a centralized digital platform that allows students from various institutions to access academic content such as syllabi, notes, notices, and old question papers in an organized manner.
- To simplify and streamline the distribution of official notices and academic updates through a dedicated and reliable communication channel.
- To enhance academic self-learning by providing easy access to credible and categorized educational resources beyond institutional boundaries.
- To implement a user friendly and responsive interface for seamless navigation and usability across different devices.
- To ensure secure user registration and content access through an admin validation mechanism.
- To promote collaboration, reduce dependency on informal platforms, and support students in preparing for exams and academic activities more effectively.

4. Scope

This student portal project aims to develop a centralized academic platform that facilitates access to verified educational resources for students across multiple institutions. The scope of the system is defined by the following boundaries:

- **Target Users:** The system will support three user roles: students, contributors, and administrators. It is intended for users from various academic institutions.
- **Academic Resources:** The portal will provide access to syllabi, lecture notes, past question papers, and academic notices. Nonacademic or multimedia content is out of scope.
- **User Access and Roles:** Registered students can access content, verified contributors can upload materials, and admins will manage verification and approvals. Advanced login methods are excluded.
- **Announcements and Notifications:** Admins can post academic announcements and updates. Messaging and real time communication features are not included.
- **Platform Accessibility:** The portal will be web based and accessible via desktop and mobile browsers. A native mobile app is not part of this phase.

5. Methodology

The development of Sikshya Kendra will follow the incremental design model, ensuring that the system is developed in a structured yet adaptable manner. This model involves building the project in successive segments or "increments," where each segment includes planning, designing, developing, and testing a set of core features. It allows the development team to gather user feedback early, manage changing requirements, and reduce project risk by focusing on smaller, manageable goals.

The incremental approach is especially beneficial for web based academic platforms like Sikshya Kendra, as it supports:

- Early delivery of usable modules for testing and feedback.
- Continuous refinement and enhancement based on realworld usage.
- Easier identification of feature gaps or user needs over time.

Initial Development Phase

In the first phase, the core module of Sikshya Kendra will focus on implementing fundamental features including:

- User registration and login system with admin validation.
- Upload and categorization of academic content such as notes, syllabi, notices, and past exam papers.
- User friendly dashboard for navigating and accessing educational materials.
- Basic announcement/notification system for real time academic updates.

Expected Benefits of the Incremental Approach

- **Scalability:** Easily accommodates more users and resources as the platform grows.
- **Flexibility:** Adapts to the dynamic needs of students from various academic backgrounds.
- **Enhanced User Experience:** Continuous improvements guided by feedback ensure the portal remains intuitive and useful.
- **Risk Mitigation:** Development in stages reduces the risk of system failure and allows for frequent testing.

- By adopting this methodology, Sikshya Kendra aims to deliver a robust, scalable, and student centric platform that evolves in response to the real needs of its users.

5.1 Requirement Identification

The requirements for Sikshya Kendra, a centralized academic portal, have been identified through a comprehensive analysis of educational technology trends, existing academic platforms (like Hamro CSIT), and user needs derived from informal student surveys and online research. The system is designed to support students from various academic backgrounds by offering an accessible platform to access verified syllabi, notes, old questions, and academic notices.

5.1.1 Literature Review

This literature review examines existing academic portals and educational resource platforms, exploring how digital solutions can improve content delivery, academic accessibility, and independent learning.

Numerous studies highlight the benefits of digital academic platforms in promoting learning outside the classroom environment. According to research, centralized repositories of academic materials enhance student performance by increasing access to verified resources and reducing dependency on informal sources (e.g., chat groups or social media). Platforms such as university portals and faculty websites help distribute course materials, old questions, and notices, but are often limited to institution specific users.

The growing need for cross institutional access to study materials has fueled the demand for generic portals that aggregate and standardize content for a broader student population. Studies further suggest that platforms that allow user contributions with moderation (admin verification) increase engagement, foster collaboration, and contribute to educational equity.

Security, usability, and mobile accessibility are frequently cited challenges in such systems. Ensuring content reliability, protecting user data, and maintaining platform uptime are critical for success. Furthermore, accessibility across devices and browsers is a key usability concern for diverse user groups.

These insights serve as guiding principles in the development of Sikshya Kendra, aiming to bridge content access gaps in Nepal's academic ecosystem.

5.1.2 Study of Existing Systems

To guide the development of Sikshya Kendra, we examined several existing educational platforms in Nepal. These platforms helped identify current gaps and inspired potential features for a more inclusive and scalable solution.

5.1.2.1. Hamro CSIT (*Hamro Csit, 2025*)

- **Overview:** Hamro CSIT is a well known online portal catering specifically to students enrolled in the CSIT program under Tribhuvan University.
- **Purpose:** It serves as a centralized resource for course content, old question papers, exam notices, and academic articles.
- **Limitations:** The platform is restricted to CSIT students and lacks support for other disciplines. Additionally, there is no provision for admin controlled content uploads or user validation mechanisms.

5.1.2.2. Mero School (*Mero School, 2025*)

- **Overview:** Mero School is an e-learning platform targeting school level students in Nepal.
- **Purpose:** It offers video based learning materials, primarily focusing on paid courses.
- **Limitations:** The commercial nature of the platform limits free content availability. It also lacks options for user contributed resources and focuses mainly on video content, reducing textual or document based learning diversity.

5.1.2.3. Saral Notes (*Saral Notes, 2025*)

- **Overview:** Saral Notes is a content sharing website that provides academic materials such as old questions, syllabi, and handwritten notes.
- **Purpose:** It aims to simplify access to academic resources for students in Nepal.

- **Limitations:** The platform suffers from weak content organization and limited user interface optimization. Furthermore, it does not support structured user validation or scalability features.

5.1.2.4. CSIT Tutor (*CSIT Tutor, 2025*)

- **Overview:** CSIT Tutor is a niche platform focused on Tribhuvan University's CSIT program.
- **Purpose:** It provides categorized resources such as notes, multiple choice questions, and old exam collections to support exam preparation.
- **Limitations:** Despite its organized interface, its scope is confined to CSIT students only and lacks cross discipline adaptability.

5.1.3 Requirement Analysis

i. Functional Requirements

- User registration and login with admin validation for content uploaders.
- Uploading and categorization of:
 - Syllabus files
 - Notes
 - Old question papers
 - Academic notices
- Search and filter features for easy navigation of resources.
- Dashboard for students to access the latest content.
- Notification system for updates or new uploads.
- Content management system for admins to approve, update, or delete resources.
- Feedback/contact form for user inquiries or content suggestions.

ii. Non-Functional Requirements

- Responsive UI across desktop and mobile devices.
- Fast content delivery and low latency page loading.
- Secure user authentication and role based access control.
- Compatibility with major web browsers.
- Backup and recovery for uploaded content and user data.

- Scalable backend architecture to support increasing user base and content volume.
- SEO optimization for public discoverability of content.

iii. **Tools and Technologies**

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Django (Python)
- **Database:** SQLite (development), MySQL (production)

5.2 Feasibility Study

A feasibility study was conducted to evaluate the potential success, risks, and viability of Sikshya Kendra.

i. Technical Feasibility

The proposed system will be developed using open source technologies that are compatible with the team's skillset and do not require costly licenses. Django, SQLite, and MySQL offer robust frameworks and database support for secure and scalable web development. The system will be designed for future scalability and mobile responsiveness.

ii. Operational Feasibility

The platform is user centric, requiring minimal training due to its intuitive design. Since it is not institution bound, users can access the platform independently. Admins will be responsible for validating user uploads to ensure content quality. A brief guide and help section will aid new users.

iii. Economic Feasibility

Since Sikshya Kendra is developed as a final year academic project using open source tools, the initial cost is minimal. If deployed publicly, hosting, domain, and basic server infrastructure will be the primary costs. Future monetization strategies (ads, partnerships, freemium features) can ensure sustainability if the platform expands.

iv. Legal Feasibility

All user data will be handled with transparency and stored securely. Proper disclaimers and consent mechanisms will be implemented. Only verified users will be allowed to upload content to avoid copyright infringement. Content moderation policies and terms of service will be established to comply with legal and ethical standards.

v. Schedule

A Gantt chart was developed to define the timeline for the Sikshya Kendra project, highlighting all critical milestones including requirement analysis, system design, hardware integration, backend and frontend development, testing, and final deployment.

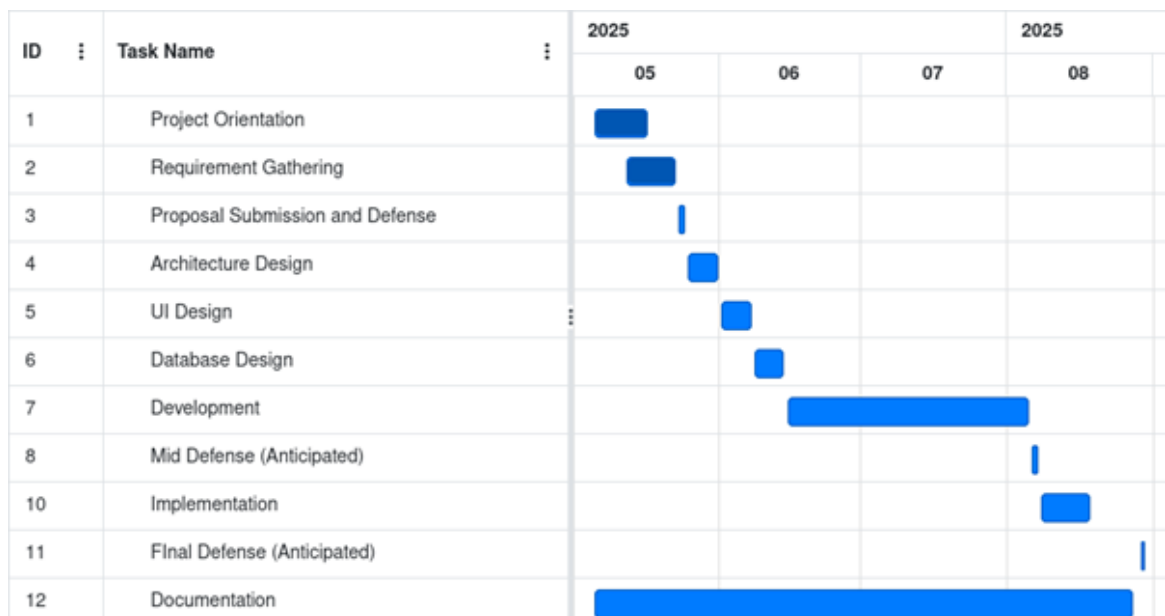


Figure i: Gantt Chart

5.3 High Level design of system

5.3.1 Use case diagram

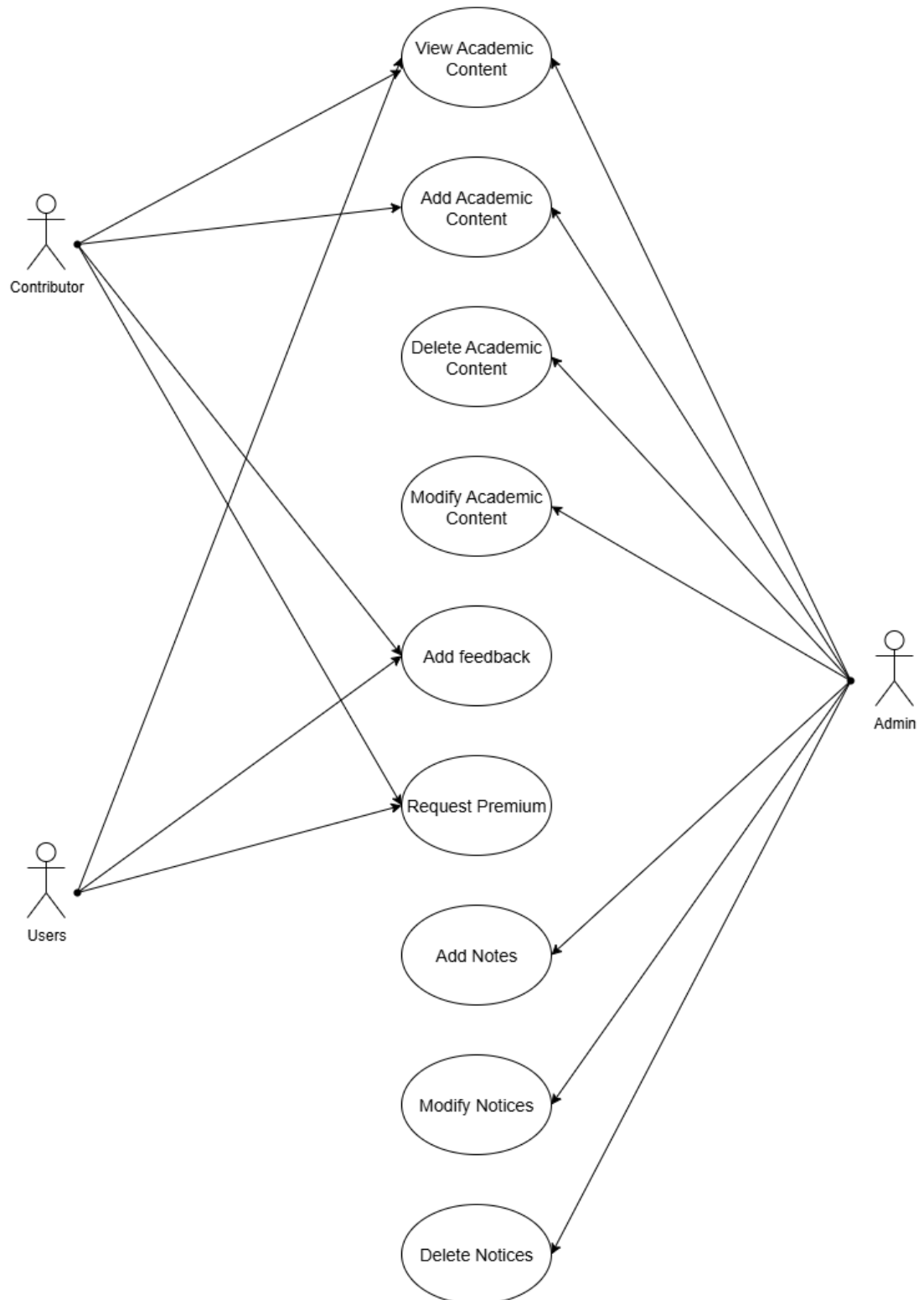


Figure ii: Use Case Diagram

5.3.2 System Flow Diagram

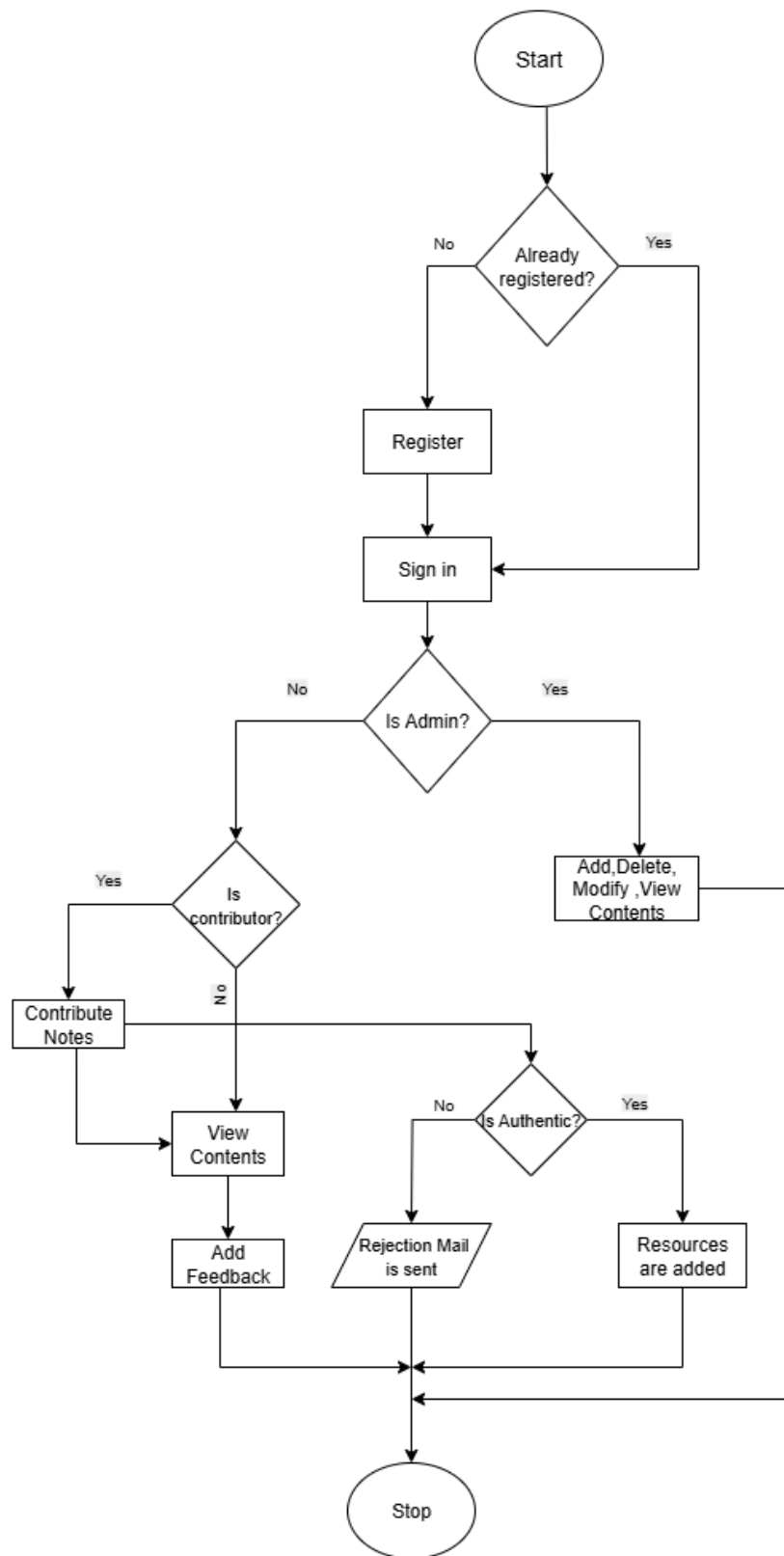


Figure iii: System Flow Diagram

6. Expected Outcomes

Upon successful completion and deployment of the Sikshya Kendra student portal, the following outcomes are anticipated:

- **Student Registration & Validation**

Students will be able to register on the platform. Admins will validate these registrations before granting full access, ensuring only verified users can contribute or download resources.

- **Academic Resource Upload**

Admins and verified contributors will have the ability to upload academic materials such as syllabi, notes, old question papers, and notices. Resources will be systematically categorized by faculty, subject, and semester.

- **Structured Academic Repository**

The portal will offer a centralized, organized repository where users can browse, search, and filter educational content. This promotes self-paced and cross disciplinary learning.

- **Notice & Announcement System**

A dedicated section will allow admins and contributors to post notices, updates, and academic announcements such as exam schedules and deadlines.

- **Responsive and Accessible Design**

The system will be mobile responsive and compatible across various web browsers and devices, making it accessible to users regardless of location or platform.

- **Admin Dashboard**

A backend dashboard will empower admins to manage users, approve or reject resource uploads, track portal activity, and remove outdated or irrelevant content.

- **Search & Navigation Features**

The inclusion of a search bar and filter options will allow users to easily locate resources based on semester, subject, or content type.

- **Scalable Infrastructure**

The platform will be built with scalability in mind, allowing for increased traffic, additional resource storage, and future integration of features like discussion forums, user ratings, and feedback mechanisms.

7. References

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